NOVA IMS INFORMATION MANAGEMENT SCHOOL

02

What is a model? Evaluation Methods and Metrics

Data Mining II| MGI 2526

February - Week 2

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Agenda

00

Introduction to Regression and Classification RECAP

01

What are models?

02

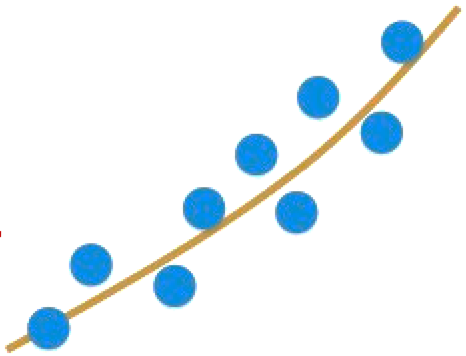
Evaluation Methods

0. Introduction to Regression and Classification

Supervised Learning

Recap from previous class

Regression

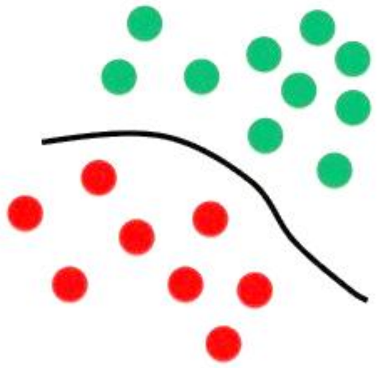


Linear Regression

Outcome variable is numerical

VS

Classification



Logistic Regression

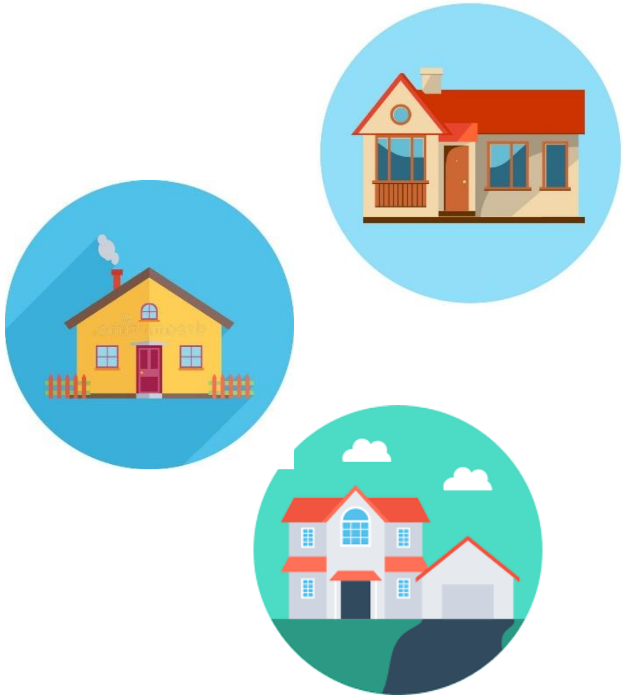
Outcome variable is categorical

Don’t make confusion with Linear Regression and Logistic Regression – Those are different concepts!

0. Introduction to Regression and Classification

Recap from previous class

Predicting the price of a house



Area	Typology	Localization	Garage	Pollution Index	Price
75	T2	Alfama	No	1	234 000
86	T1	Campolide	Yes	2	210 000
93	T2	Santos	No	2	254 000
134	T4	Arroios	No	3	457 000
96	T3	Alfama	No	1	367 000
54	T2	Alcântara	No	2	219 000
167	T5	Restelo	Yes	2	675 000
95	T2	Arroios	Yes	3	?

0. Introduction to Regression and Classification

Recap from previous class

Predicting the probability of someone having diabetes

ID	Gender	Weight	Height	Pregnancies	Diabetes
12562	M	78	175	0	0
62735	F	66	155	3	1
84638	F	91	165	1	1
1232	M	89	187	0	0
13738	M	101	172	0	1
3274	M	81	179	0	0
3283	F	72	169	0	0

3627	F	93	185	0	?

0. Introduction to Regression and Classification

Predictors, Independent variables, Explanatory variables, Input variables, …

Target, Outcome, Output, Dependent variable, …

Area	Typology	Localization	Garage	Pollution Index	Price
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Independent Variables » Dependent Variable

Relationship ?

Explain

Predict

01. What are Models?

The output of machine learning algorithms run on data.

1. What are models?

Introduction

Machine learning techniques that learn patterns from untagged data.

1. What are models?

Introduction

- As in any other computer tasks, modeling requires a “program” providing detailed instructions
- These instructions are typically mathematical equations, which characterize the relationship between inputs and outputs
- Formulating these equations is the central problem in modeling

1. What are models?

Introduction

Models
How can we define these instructions and equations?

- **Fixed Models**
- **Parametric models**
  Model driven
- **Non-parametric models**
  Data driven

1.1. What are models?

Fixed Models

- Closed-form equations that define how the outputs are derived from the inputs
- Being all the characteristics fixed when the equations are derived we refer them as fixed models
- Suitable for simple and fully understood problems

Example:

Compute how much it takes an apple to hit the ground on Earth:

$$t = \sqrt{\frac{2h}{9.8}}$$

Most problems are too complex and / or not sufficiently understood for us to use fixed models.

1.2. What are models?

Parametric Models

- You know enough about the problem to define the general relationship between inputs and outputs
- But some parameters are unspecified – those are chosen by examining data examples

Example:

Computes how much it takes an apple to hit the ground on Mars:

$$t = \sqrt{\frac{2h}{g}}$$

The rate of acceleration

You need to estimate the parameter g!

1.2. What are models?

Parametric Models

You need to estimate the parameter g!

Go to Mars and collect some data:

Height (h)	Falling time (t)
0.5	0.2
1.3	0.4
2.8	0.46
4	0.68
7.3	0.7

Inputs
(in this case the input vector has only one component)

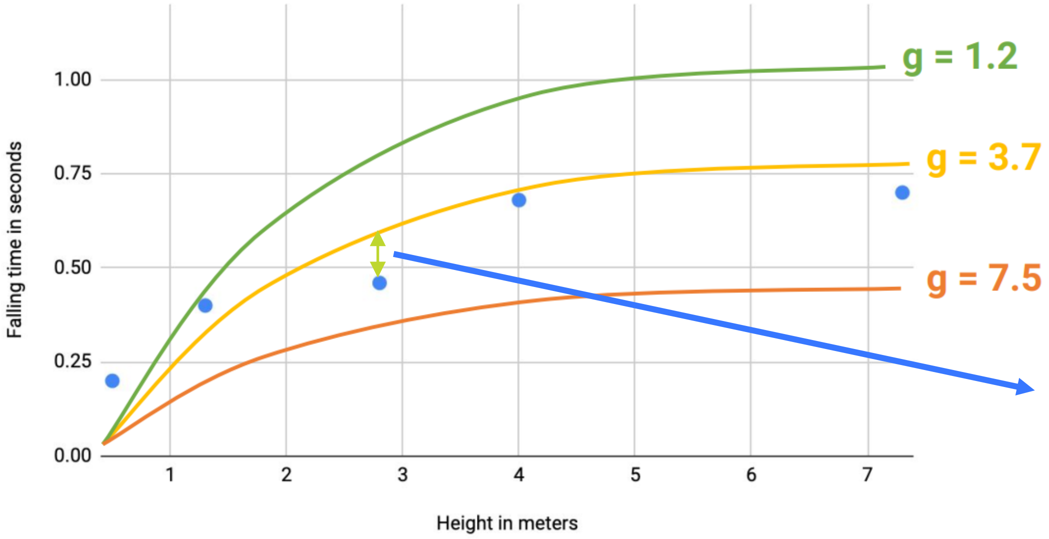
Outputs

1.2. What are models?

Parametric Models

A value should be selected for g so the model produces estimates close to the measured times when presented with the corresponding heights as input.

Search for the parameter that leads to the predictions that are closest to the observed data:



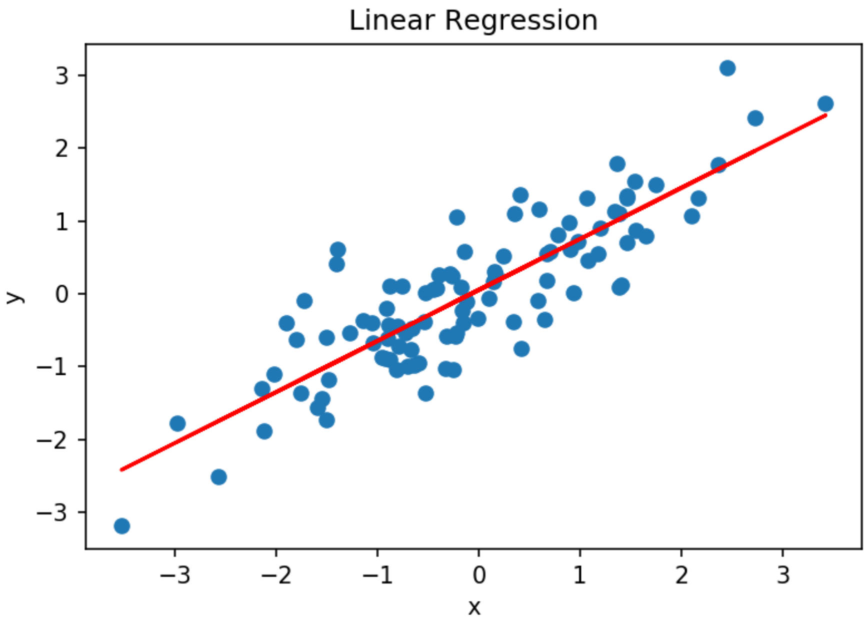
Seems to be the best model

Due to inaccuracies in the structure of the model or measurement errors

In this case is easy to define a good value for g but this is not generally possible in most problems which involve complex relationships and multiple variables

1.2. What are models?

Parametric Models

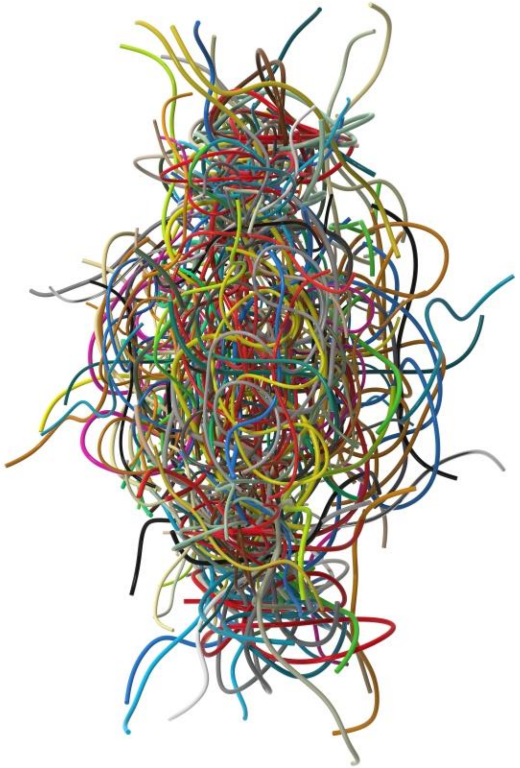


- A common example of a parametric model is the linear regression.
- The assumption is that there is a linear relationship between inputs and outputs:

$$y_i = \beta_0 + \beta_1 x_{i1} + \ldots + \beta_k x_{ik} + \varepsilon_i, i = 1, \ldots n$$

1.3. What are models?

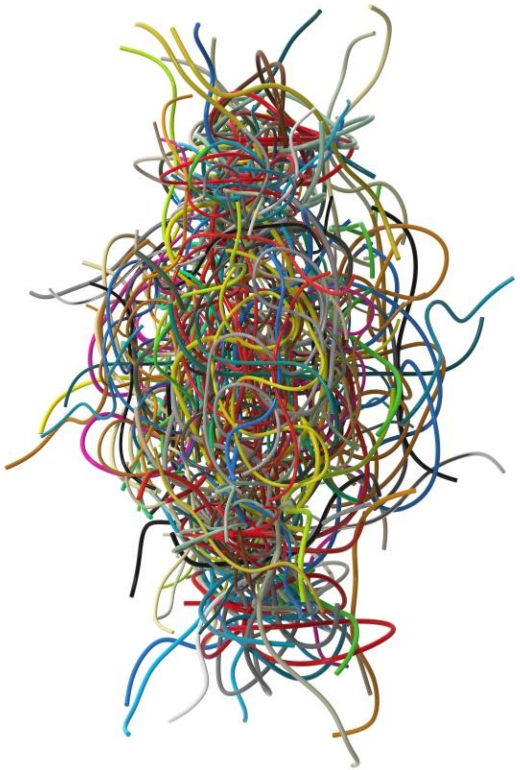
Non-parametric Models



- Models that rely heavily on data, instead of human expertise, can be called data-driven models
- Used in complex problems where the relationship between inputs and outputs is not known
- Requires large amounts of data

1.3. What are models?

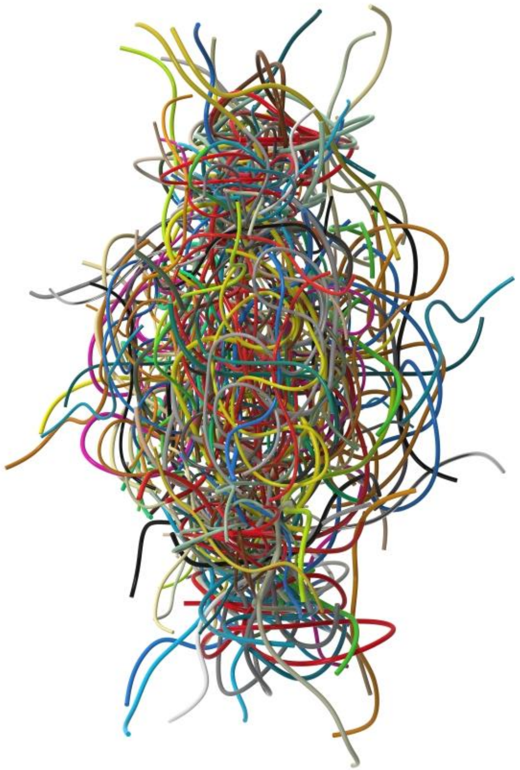
Non-parametric Models



- The basic premise is that relations consistently occurring in the dataset will recur in future observations
- One important benefit is that non-parametric methods do not require a thorough understanding of the problem
- Models can be arbitrarily complex
- Closer to inductive reasoning

1.3. What are models?

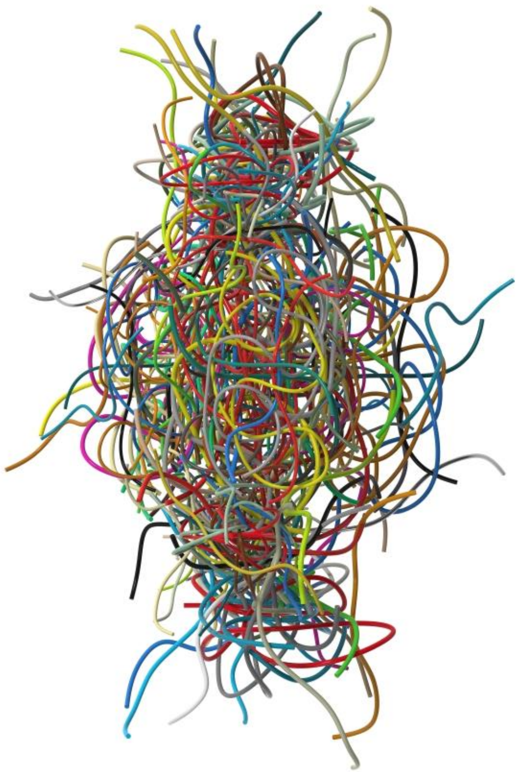
Non-parametric Models

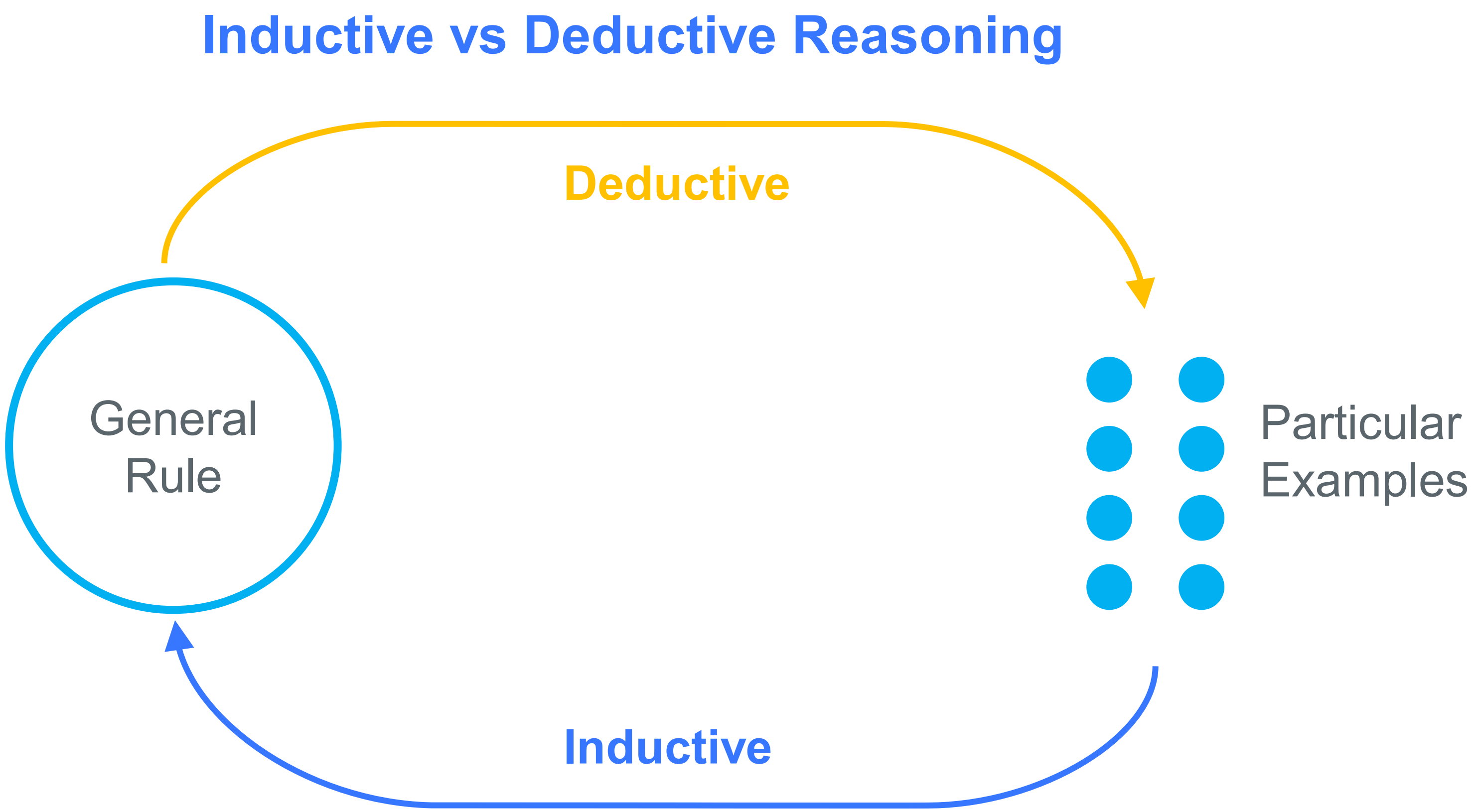


- In real use-cases, most problems are too complex to use a fixed or parametric model
- Thus, non-parametric models are the most used
- However, you should always start with the simplest model and only increase complexity when the problem requires it!

1.4. What are models?

Parametric vs. Non-parametric Models



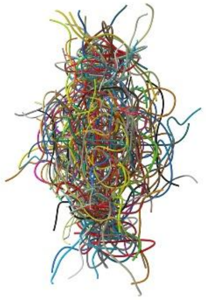


1.4. What are models?

Parametric vs. Non-parametric Models

Parametric model

“A learning model that summarizes data with a set of parameters of fixed size (independent of the number of training examples) is called a parametric model. No matter how much data you throw at a parametric model, it won’t change its mind about how many parameters it needs.”



Nonparametric model

“Nonparametric methods are good when you have a lot of data and no prior knowledge, and when you don’t want to worry too much about choosing just the right features.”

Russell, Stuart J. (Stuart Jonathan). Artificial Intelligence : a Modern Approach. Upper Saddle River, N.J. :Prentice Hall, 2010.

1.4. What are models?

Parametric vs. Non-parametric Models

Parametric model | assume known distributions in the data

Simplicity – Easier to understand and interpret

Training Speed – Faster to train and learn from data

Less training data – Does not require a huge quantity of data for training and work well even if the fit to the data is not perfect

Constrained – Limited by the functional form chosen

Limited complexity – Better suited to less complex problems

Poor fit– In practice the methods are unlikely to match the underlying mapping function – do not offer the best fit to data

In pre-processing - the analyst often spends considerable time transforming data so it stands with some specific distribution (for example normal distribution)

Linear Regression

Logistic Regression

Perceptron

Naïve Bayes

…

1.4. What are models?

Parametric vs. Non-parametric Models

Nonparametric model | do not assume distributions in the data

Flexibility – Capable of fitting a large number of functional forms

Power – No assumptions (or weak ones) about the underlying function. Learn from data.

Performance – Can result in higher accuracy since they offer better fit

In pre-processing – there is no distribution assumptions and time can be saved in preprocessing steps (e.g. no need to transform the data to normal distributions)

Training data – Require more data than the parametric models

Slower – Slower due to the several parameters needed to train

Overfitting – Higher risk of overfitting the training data

Lack of interpretability – Harder to explain why specific predictions are made in some of the algorithms

KNN

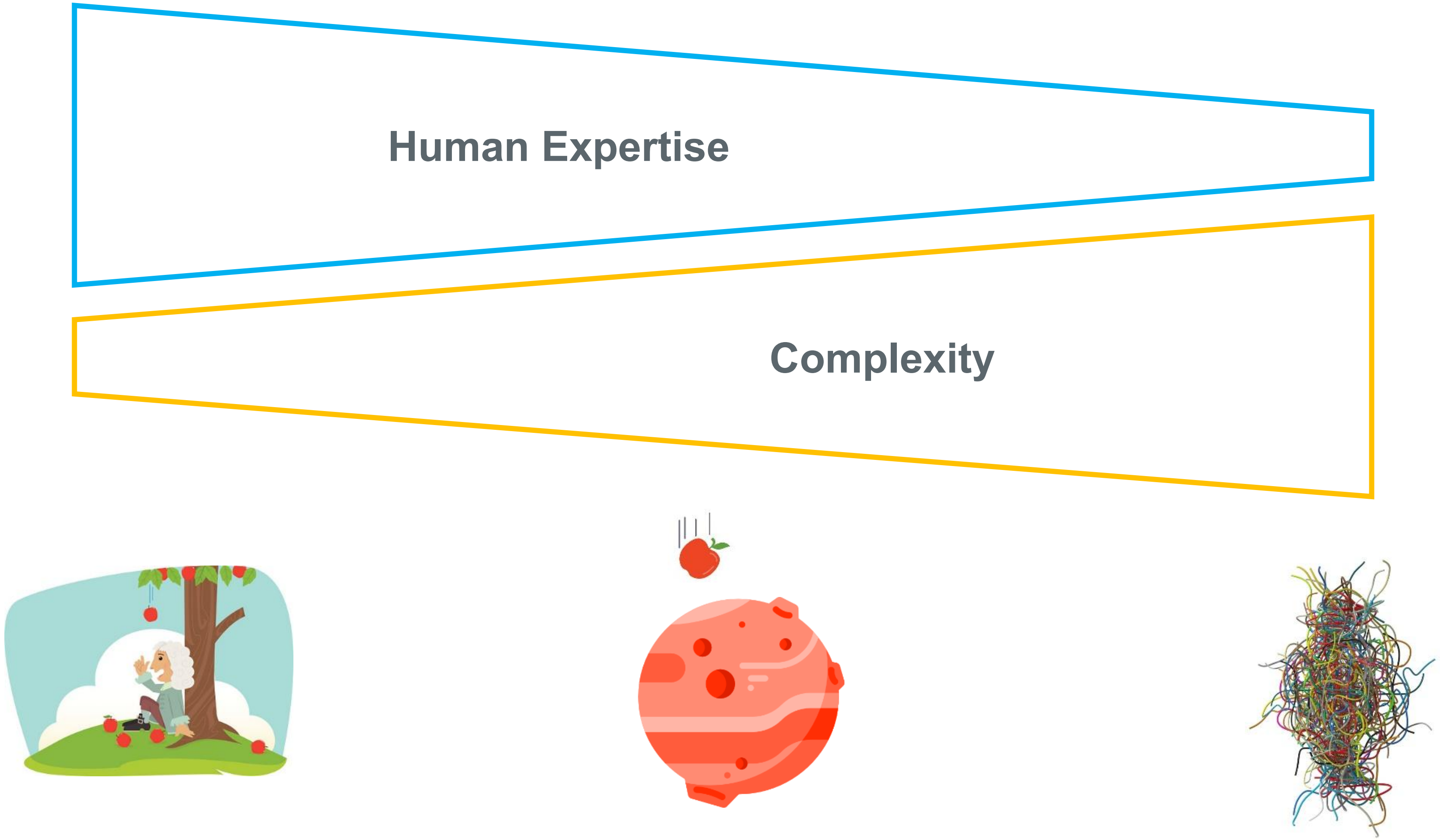
Decision Trees

Non-linear SVM

...

1. What are models?

Models RECAP



1.5. What are models?

When building models...

- Which is the best algorithm to use?
  - The concept of **no-free lunch theorem**
- What is the right number of attributes to use?
  - The concept of **complexity**
- Is this problem a separable one?
  - The concept of **separability**
- Too much learning?
  - The concept of **overfitting**

1.5.1. What are models?

The No Free Lunch Theorem

"If an algorithm performs better than random search on some class of problems then it must perform worse than random search on the remaining problems."

David Wolpert and William G. Macready

In: No Free Lunch Theorems for Optimization

https://ti.arc.nasa.gov/m/profile/dhw/papers/78.pdf

- No single machine learning algorithm is universally the best-performing algorithm for all problems!
- You need to understand the trade-offs

1.5.2. What are models?

The problem of complexity

- There are problems that are too complex to deal with
- Sometimes it is not feasible to collect enough data and to allow enough processing time to generate an optimal model
- Choosing the right features is a way of simplifying the problem and reducing the “search space”
  - Important attributes vs redundant attributes: which are the variables that are relevant for the task?

1.5.2. What are models?

The problem of complexity

What is the right number of features to use?

Small number of attributes

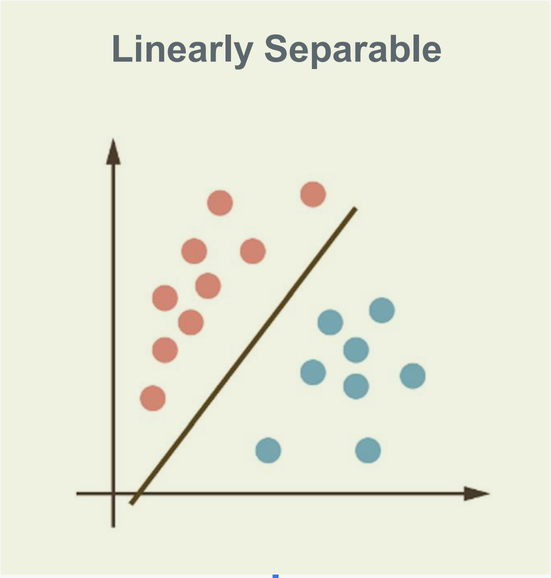
- Unable to distinguish between classes

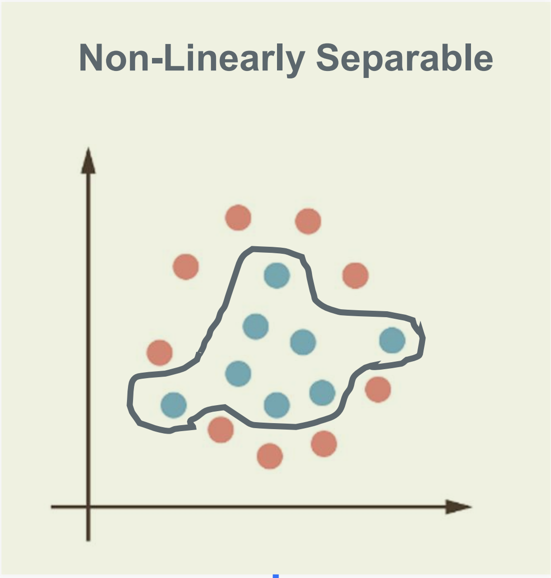
Large number of attributes

- The most usual case
- The curse of dimensionality
  - As the number of dimensions increases the space gets sparser and finding groups is more difficult
- Hard to visualize and some “strange” effects

1.5.3. What are models?

The problem of Separability





It's possible to have no errors

Will always have errors

1.5.4. What are models?

The problem of overfitting

Models rely on training data to learn

If we allow too much complexity, the model will “memorize” the training data, instead of extracting useful relationships

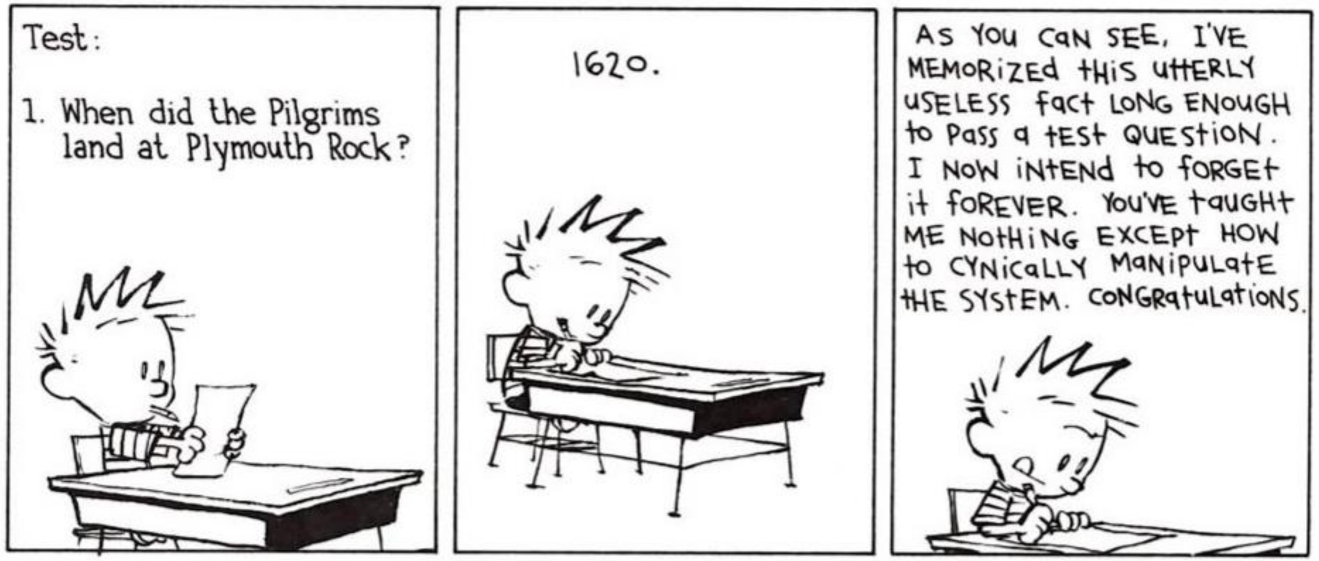
OVERFITTING

1.5.4. What are models?

The problem of overfitting

Memorizing vs Understanding

- Overfitting is like when someone memorizes things to pass an exam
  - He'll be too biased on the exercises he saw in classes
  - If he gets a slightly different question in the exam, he won't know how to answer

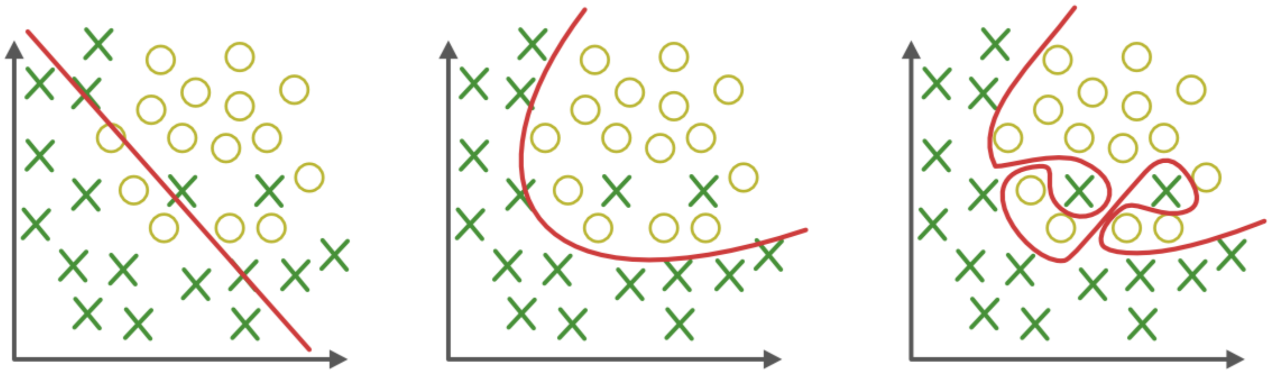


1.5.4. What are models?

The problem of overfitting

Underfitting vs Overfitting

In a classification problem

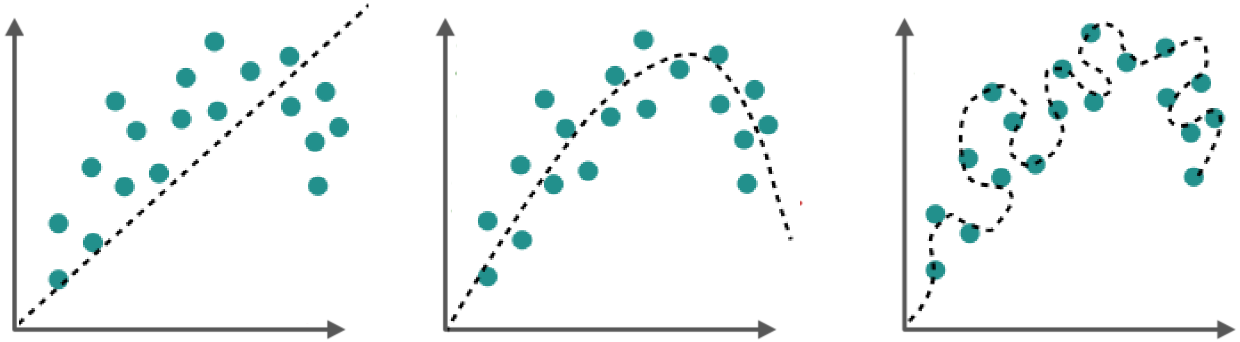


Underfitting
(Too simple to explain the variance)

Appropriate fit

Overfitting
(Forcing the fit! Too good to be true)

In a regression problem



02. Evaluation Methods

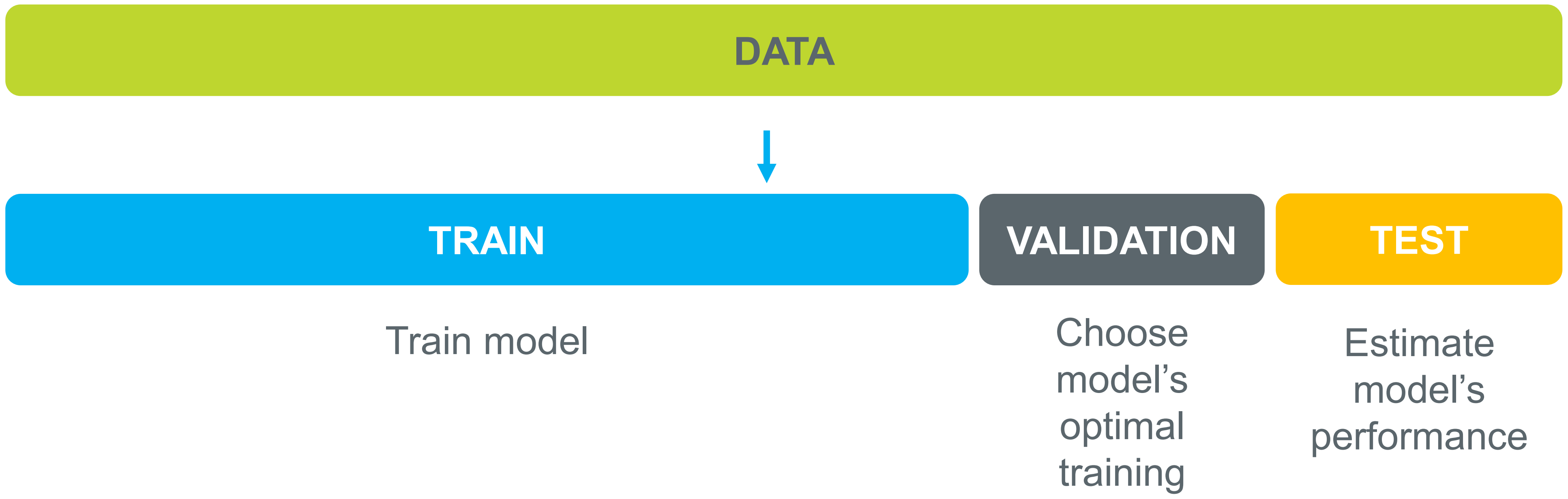
2. Model selection approaches

The concept of overfitting

How to avoid overfitting?

How to prepare for the unknown?

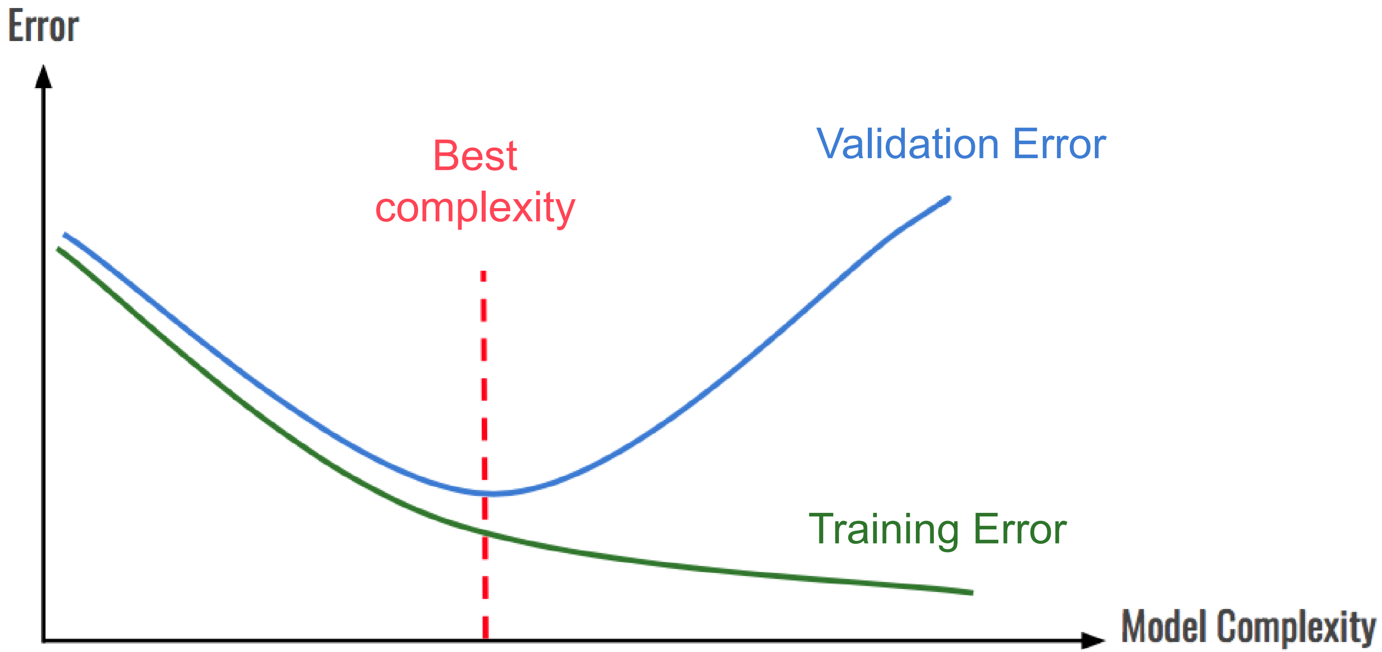
- Keep some data aside!



2. Model selection approaches

The concept of overfitting

How to avoid overfitting?

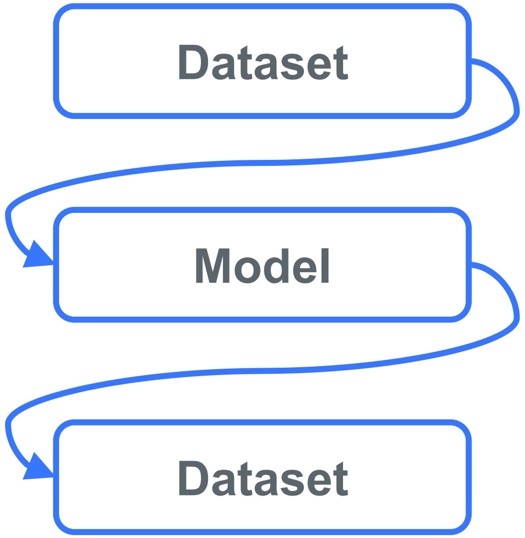


2. Model selection approaches

How to split your data?

TRAINING SET	The bigger the better the classifier.
VALIDATION SET	The bigger the best the estimate of the optimal training.
TEST SET	The bigger the best the estimate of the performance of the classifier on unseen data.

2. Model selection approaches



We can estimate and evaluate the performance of the model on the training data

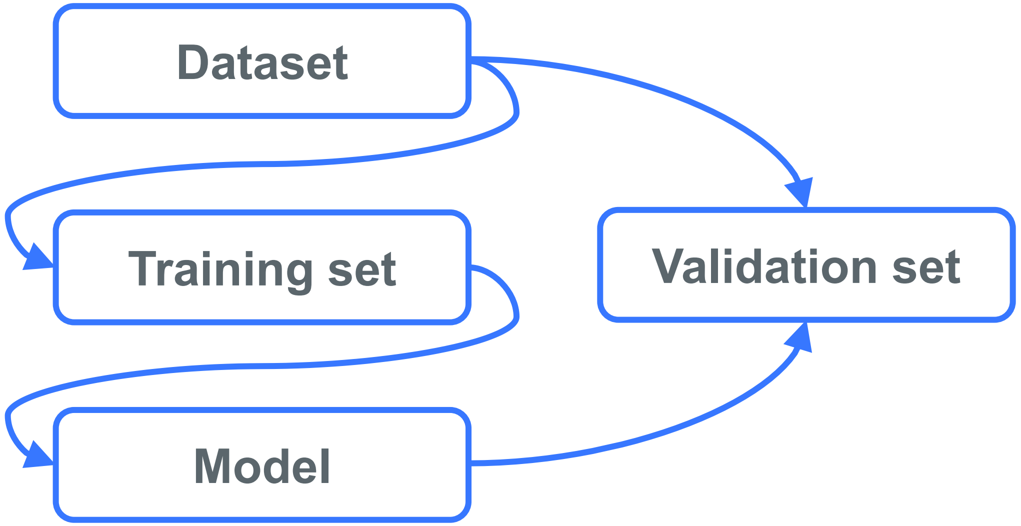
- However, estimates based only on the training data are not good indicators of the performance of the model on unseen data
- New data will probably not be exactly as the same as the training data – these estimates suffer from overfitting

This is not the ideal!

2.1. Model selection approaches

Hold-out Method

One possible solution:



- Use an independent validation set
- Used when:
  - There is plenty of data

RULE OF THUMB

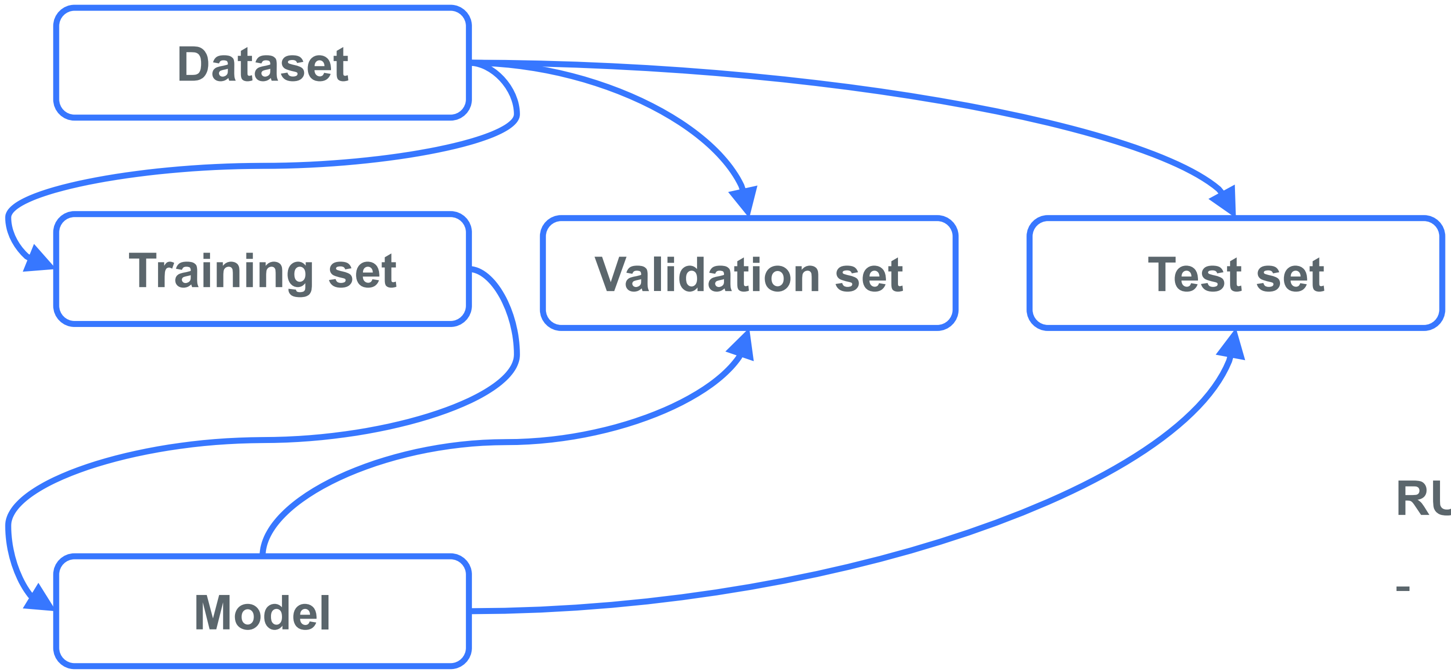
- 70 % for training set
- 30 % for validation set

(Some authors defend 80% train and 20% validation – it depends on the size of the dataset)

2.1. Model selection approaches

Hold-out Method

When there is plenty of data we can split our dataset into training, validation and test set:



RULE OF THUMB

- 70 % for training set
- 15 % for validation set
- 15 % for test set

2.1. Model selection approaches

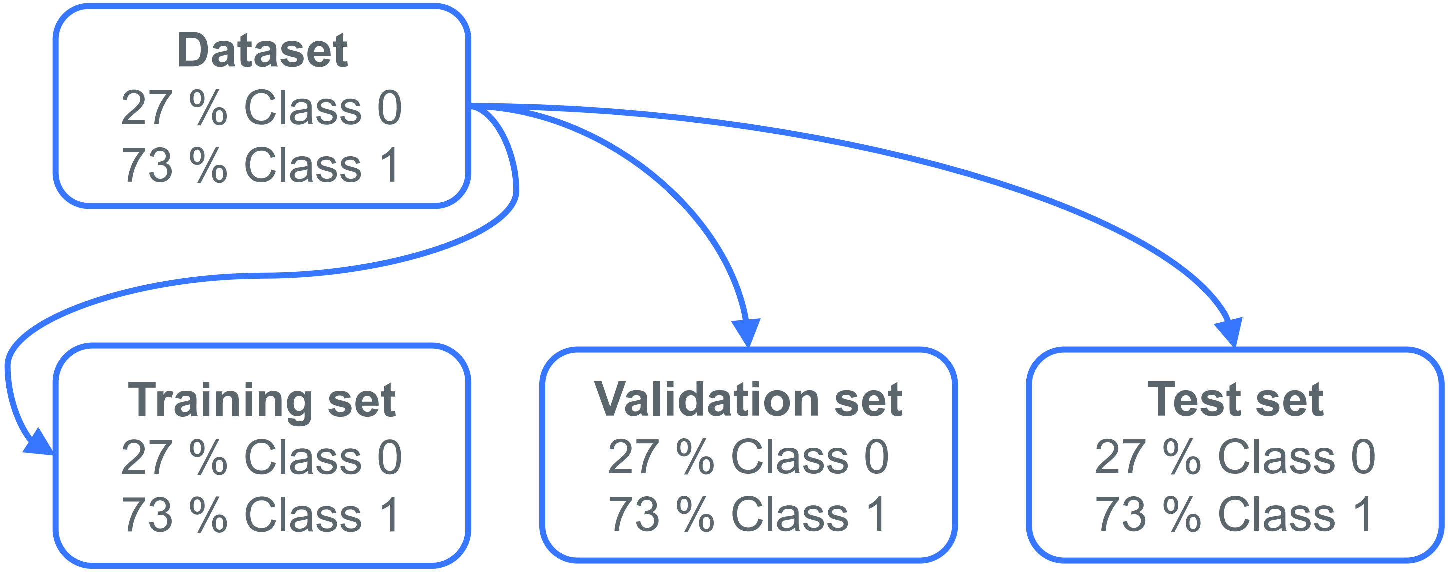
Question: What if your dataset is imbalanced?

Answer: Then, for some runs, certain classes may be not (well) represented. We need to assure that the data partition will not change the original proportions of each class on the new datasets.

⬇

Stratified sampling

Sample each class independently such that each dataset has the same ratio of any given class



2.1. Model selection approaches

Hold-out Method

PROBLEMS OF THE HOLD-OUT METHOD

- It only provides a single estimate
    - **SOLUTION** : Use a repeated hold-out method
- It requires a large enough dataset
    - **SOLUTION** : Use K-Fold cross-validation

Repeated hold-out method

1. Run the holdout method many times
2. Each run will have a different train and validation set
3. Each run will lead to slightly different metric values

↓

Average value allows a more robust estimate
Compute standard deviation to estimate variability

Still not optimum
The different tests sets overlap, but we would like all our instances from the data to be tested at least once
Can we prevent overlapping?

2.1. Model selection approaches

Hold-out Method

PROBLEMS OF THE HOLD-OUT METHOD

- It only provides a single estimate
  - SOLUTION : Use a repeated hold-out method
- It requires a large enough dataset
  - **SOLUTION :** Use K-Fold cross-validation

2.2. Model selection approaches

K-Fold Cross Validation

	DATA				
1st Iteration	VAL	TRAIN	TRAIN	TRAIN	TRAIN
2nd Iteration	TRAIN	VAL	TRAIN	TRAIN	TRAIN
3rd Iteration	TRAIN	TRAIN	VAL	TRAIN	TRAIN
4th Iteration	TRAIN	TRAIN	TRAIN	VAL	TRAIN
5th Iteration	TRAIN	TRAIN	TRAIN	TRAIN	VAL

Step 1

Split data into k subsets of equal sizes

Step 2

Each turn, use one subset for testing / validation and the remainder for training

Step 3

Average all estimates to get a final value

The common pipeline…

After tuning the model using cross-validation, you evaluate the model's performance on a completely separate test set that was never used during cross-validation

- This ensures that the final performance metric reflects how the model generalizes to unseen data.

2.2. Model selection approaches

K-Fold Cross Validation

SOME TIPS

- Use a **stratified sample** (it reduces the variance)
- **10 folds is by far the most used**
    - Extensive experiments have shown that this is the best choice to get an accurate estimate
    - Smaller datasets can demand a small number of folds
    - Big datasets can make use of more folds (and in that way get a more robust estimate)

2.3. Model selection approaches

Other options

- Use a **repeated stratified cross-validation** to further reduce variance
- **Leave–One–Out Cross-Validation:**
  - CV with as many folds as instances
  - Very expensive computationally
  - Makes best use of the data

THE BEST METHOD DEPENDS ON EACH CASE !

2.4. Model selection approaches

Compare two models

Step 1 Choose your **metric**

Step 2 Choose an **evaluation method**

Step 3 **Run** the evaluation for both models and collect metrics

Step 4 **Compare** metrics

Step 5 **Pick** the model with the best performance

How to be sure one model is always better than the other?

Using different datasets will lead to slightly different results

The train and validation on different splits and the computation of standard deviation will help you

(you can even go further and use a hypothesis test to prove or disprove measurable hypothesis)

2.4. Model selection approaches

Compare two models

What is the best model?

- The **simplest** representation of knowledge
  - Easier to understand
  - Decision Trees vs Neural Networks
- The knowledge representation with **less error**
- The most simple…
  - **Occam's Razor** – " the simplest explanation is usually the right one"

2. Model selection approaches

Wrap-up

- Use **test sets** and the hold-out method for “large” data
- Use the **cross-validation** method for middle-sized data
- Use the **leave-one-out** method for small data
- Take advantage of **stratified sampling**
- **Don’t use test data for parameter tuning** – use separate validation data

Obrigada!

Thank you!

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